QR CODE **WATCH THE VIDEO BEFORE YOU SOLVE** YouTube • Lesson 11

**RULE · Composition (kompozitsiya): $f(g(x))$**

Evaluate inside-out: compute $g$ first, then plug the result into $f$.

$f(x) = 3x - 2,\ g(x) = x^2 + 1$: $\quad f(g(2)) = f(5) = 13$

*$f(g(x)) \neq g(f(x))$ in general – order matters.*

**RULE · Vertical transformations**

| Form | Effect |
|---|---|
| $f(x) + k$ | Shifts graph up $k$ (down if $k < 0$) – output changes, not input |
| $-f(x)$ | Reflects graph over the $x$-axis |

$f(x) = x^2 - 4$: $\quad f(3) + 5 = 5 + 5 = 10,\quad -f(3) = -5$

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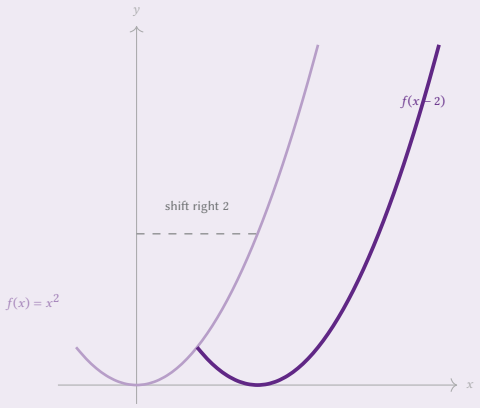
**RULE · Horizontal transformations**

| Form | Effect |
|---|---|
| $f(x+h)$ | Shifts graph LEFT $h$ |
| $f(x-h)$ | Shifts graph RIGHT $h$ – opposite of the sign |
| $f(-x)$ | Reflects graph over the $y$-axis |

$f(x) = x^2 + 2x$: $f(x-2)$ at $x = 5$ is $f(3) = 15$; $f(-x)$ at $x = 3$ is $f(-3) = 3$




*Direction inside the parentheses is always reversed from what it looks like.*